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Applying a Duchon Spline Based on a Gauss-Newton Optimizer for Coupled Flow and Geomechanics for CO₂ Sequestration

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Abstract

We utilize a superposition principle and an inversion approach to compute a consistent stress field that honors the available measurements and equilibrium. We address the compatibility problem by minimizing an objective function, which allows us to tune the degrees of freedom (DOF) we represent by piecewise linears as a depth function, i.e., like well-logs. As an extension, we propose an object-oriented software architecture that relies on an abstract "evaluator," injecting the variation of the underlined properties into existing finite element codes with minimal modifications. We explain the architecture, its testing, and applications to synthetic but realistic reservoir models for CO₂ capture and sequestration applications (CCS).

We present comprehensive numerical results encompassing the whole 3-D synthetic field heterogeneous poroelastic cases, one-way coupled with our in-house reservoir simulator, emphasizing CCS. We also tackle a gigantic synthetic Middle Eastern reservoir to show the approach's scalability on cluster computers. We present tabular information, such as the number of iterations, objective function values, and flops, to assess the optimization process performance. The results dictate that our optimization-based approach and evaluator's architecture can handle realistic stress fields that match the predefined measurements with a minimal computational cost regarding iterations and simulation jobs.

Objectives

Field-level reservoir compaction and subsidence CO₂ storage applications (CCS) require computing an initial stress field that must equilibrate itself. There may also be measurements we want to honor. Given the underlying uncertainties, computing the virgin stress field is complicated. In a previous SPE paper, we proposed an optimization-based approach to tackle the above compatibility problem [1]. We follow up by presenting additional implementation details, such as the proper parametrization of the mechanics' problem, emphasizing carbon capture and sequestration in this paper.

Background

The literature lacks guidance on systematically computing a compatible virgin stress field. Authors claim that mechanical properties and in-situ stresses should self-equilibrate, which is not valid for most practical applications. Also, the resulting displacement fields could exhibit larger-than-acceptable magnitudes. The present proposal details the implementation and software architecture to overcome these gaps and limitations. It paves the way for existing software to tackle the above initialization as a given minimization, i.e., a manageable parametrized nonlinear least-squares problem [1-4].

In a previous SPE paper [1], we introduced a novel approach that employs a superposition principle and an optimization-based approach to compute a consistent stress field that honors the given measurements and equilibrium. We posed the compatibility problem by minimizing an objective function in the classical Bayesian framework, which allowed us to invert the unknown degrees of freedom (DOF) that we represent by piecewise linears as a function of depth, i.e., resembling well-logs in petrophysics. We also utilized a derivative-free (DFO) Gauss-Newton (GN) optimizer that requires about half the number of simulations compared to an equivalent gradient-based optimizer [2-4]. We approximate the sensitivity matrix by interpolating the resulting predictions with Duchon splines (DS), i.e., radial basis functions, instead of the well-known QR approach by base vectors (BV). We focus on implementation details in this paper, particularly the object-oriented approach to tackle the in-situ stresses and mechanical properties using the concept of a generic evaluator, which relies on the design pattern of a factory of objects.

Studies have explored the integration of CCS within enhanced geothermal systems (CPG), demonstrating the feasibility of CO₂ storage in geothermal reservoirs and highlighting the benefits of complex fracture networks for CO₂ sequestration. Leveraging geothermal heat and generated energy for CCS operations presents a promising pathway toward a more sustainable energy landscape [5,6]. Numerical simulators are crucial for simulating and optimizing CPG systems [7-9]. A massively parallel numerical simulator is particularly advantageous, as it can manage the complex requirements of CPG systems. The key benefits of using a massively parallel numerical simulator for CPG optimization include accurate and detailed simulations, scalability, and optimization capabilities. In that sense, the present work contributes to a first step towards tackling the multi-physics CCS problem. We expect to explore further thermal simulations coupling in forthcoming works.

The remainder of the paper is as follows: we introduce the deterministic problem in the next section. After that, we explain how to implement the abstract evaluator and then present two concrete numerical examples for a real CCS site and a synthetic but large Middle Eastern reservoir. Finally, we state concluding remarks and the path forward.

Deterministic poroelastic problem

We describe the poroelastic deterministic problem for the sake of completeness herein, but refer the reader to our SPE paper [1] for a complete treatment. We must first equilibrate and solve the reservoir compaction and subsidence problem, one-way coupled with a flow simulator. Toward that end, we first address the initialization problem for the virgin stress field, see Figure 1 (left), followed by a transient stage (right) where we compute variation from the initial state through handling pressure snapshots. Figure 1 highlights the pay zone as a red area in the center, while we extend the domain toward the surroundings to decrease the uncertainty in the boundary conditions (BCS). We often refer to those extensions as overburden (top), underburden (bottom), and sideburdens [1].

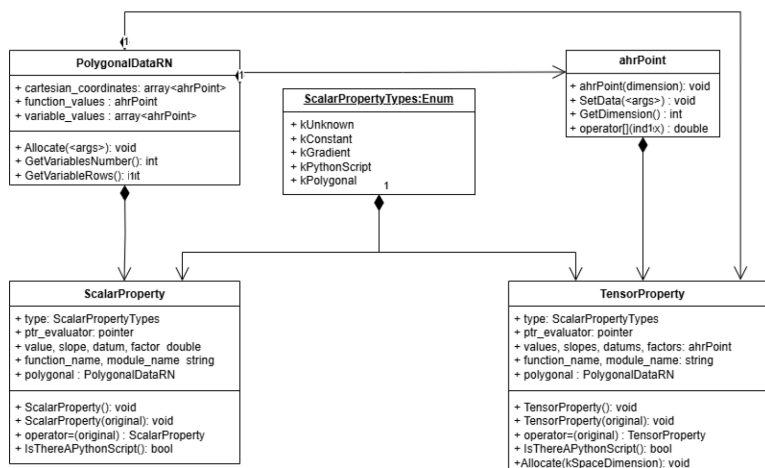


Figure 2—It depicts the UML diagrams for handling scalar and tensor properties.

The implementation proceeds by defining interfaces for abstract evaluators as shown in Figure 3. The classes "ScalarPropertyEvaluator" and "TensorPropertyEvaluator" are concrete implementations. An evaluator needs to know what evaluation is supported through the field "type." Smart shared pointers cover specific intermediate objects, such as Python scripts, Duchon splines (polygons), and constants or gradients. We also implemented specific constructors to initialize the evaluator for those use cases. The concrete method "Evaluate" returns the property at the given depth based on the given options. We wrapped up the implementation by coding a dedicated service or helper class that instantiates the intermediate objects and calls the proper evaluator based on the input arguments. Our finite element code consumes all the classes above to enforce the variable properties and boundary conditions accordingly. Boundary conditions are prescribed at preprocessing time, and properties in the volume are consumed during assembling the stiffness matrix, for instance. The implementation is pragmatic because it can be updated to support other evaluation methods besides those already included in the enumeration. We also coded in C++ unit tests in "GoogleTest" to cover our implementation. Our testing covers the data structures, the evaluators, and the service. We test for accuracy and robustness, covering regular and stringent control paths with a dozen unit tests.

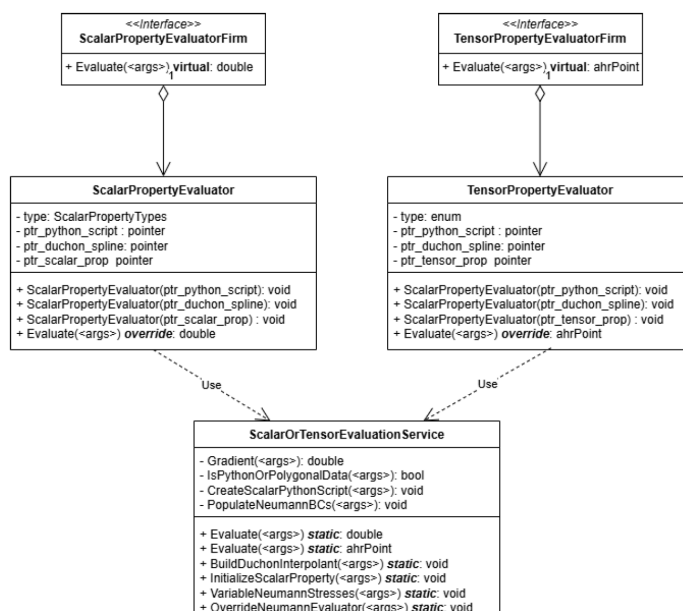


Figure 3—It shows the interfaces, concrete evaluator, and service to implement the general evaluator.

Numerical experiments

The authors employed the Integrated Parallel Finite Element Analysis program (IPFA) to solve the elasticity parametrized problem coupled by memory with an optimizer in Python (Spyder IDE, Anaconda) via the CTypes library. IPFA is a C++ application with characteristics described in references [9-13] and has been extensively benchmarked and tested for years. The examples below were run on a MacBook Pro laptop with an Intel(R) i9-9880H CPU@2.30 GHz, twelve cores, and 32 GB of RAM. We utilized Aramco's in-house geomechanics simulator coupled with the Python scripts similarly [8] in the example regarding the gigantic reservoir.

The tables below present the optimization results that summarize all runs, for which we vary the number of walkers, i.e., the number of initial points in the training repository that we utilize to solve a given optimization problem. For all tables, the first column lists the tested optimizer variant, the second column documents the number of iterations spent until convergence, the third column documents the number of objective function evaluations, i.e., a proxy to the number of flops, and the last column reports the obtained objective function value. Below are two concrete numerical examples for a real CCS site, i.e., the Cranfield model, and a synthetic but large Middle Eastern reservoir that shows scalability on cluster computers.

Coarse Cranfield model

We revisit a coarser model from the Cranfield CO₂ Storage Project that we discussed in detail [12]. We utilize the porosity and permeability fields that Figure 4 depicts. In this synthetic exercise, we aim to deplete the pay zone completely. We thus assume 64 vertical producer wells with a BHP of 3000 Psi while the initial pressure is 10000 Psi. The reservoir grid size is $47 \times 44 \times 22$, and we run a single-phase model for twenty years with a viscosity of 0.4 cp and total compressibility of $1.4 \cdot 10^{-5} \text{ Psi}^{-1}$. We employ the fully implicit scheme described in [1,9,11,12] with a constant timestep size of 15 days.

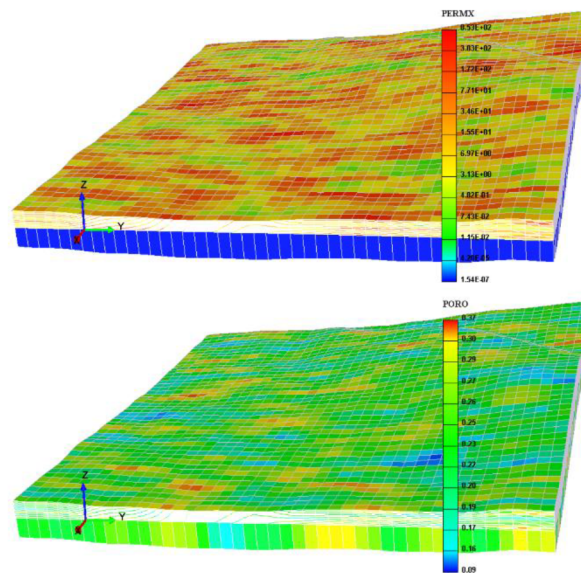


Figure 4—It shows the coarse Cranfield's permeability (top) and porosity (bottom) fields.

Our objectives with this example are twofold. First, we aim to assess the role of heterogeneity in the optimization process. Secondly, we wish to tackle a more massive, i.e., higher number of DOF, and thus a realistic case. Towards that end, we run two distinct situations, one in which we consider layered properties as before, and the second one in which Young's modulus varies as a function of the porosity according to the following correlation [14]:

$$E = E_p \cdot \left(1.0 - \frac{\phi}{0.41}\right)^{1.19}, \quad (3)$$

where the equation holds at the reservoir, where we know the porosity values, and E_p is the pay zone's level of Young's modulus.

We exported 48 pressure snapshots from the flow simulation, e.g., every 152 days, for one-way coupling purposes. Figure 5 depicts sample pressure fields at increasing times every 3.33 years, as indicated from left to right and top to bottom, spanning 20 years of simulation. There is an evident depletion scenario in which pressure drops from wells and quickly propagates into the surrounding areas. Notice that the black dots in the pictures highlight the location of wells. Once again, we derive a geomechanics mesh by reconstructing the reservoir model as explained in [1,15,16].

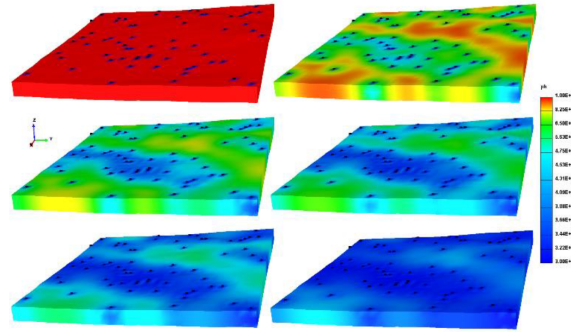


Figure 5—It depicts pressure snapshots every 3.33 years from left to right and top to bottom.

Additionally, regarding the pay zone's mesh size, we consider the following dimensions: $N_u = 4$, $N_o = 10$, and $N_c = 6$. We extrapolated the reservoir towards the side-burdens with the reconstruction approach described in [15,16]. Figure 6 shows the geomechanics mesh encompassing 118944 elements with 126540 points. Again, we wanted to honor the pay-zone mesh exactly as shown. The Figure also depicts the heterogeneous Young's modulus field correlated to the porosity. The BCS and the initial condition for mechanics were described in [1]. Regarding mechanical properties, we assume layered RS with Young's modulus $E_u = 4 \times 10^4$, $E_p = 3 \times 10^4$, $E_o = 6 \times 10^4$ [Psi] and Poisson's ratio, $\nu_u = 0.25$, $\nu_p = 0.3$, and $\nu_o = 0.3$ varies accordingly.

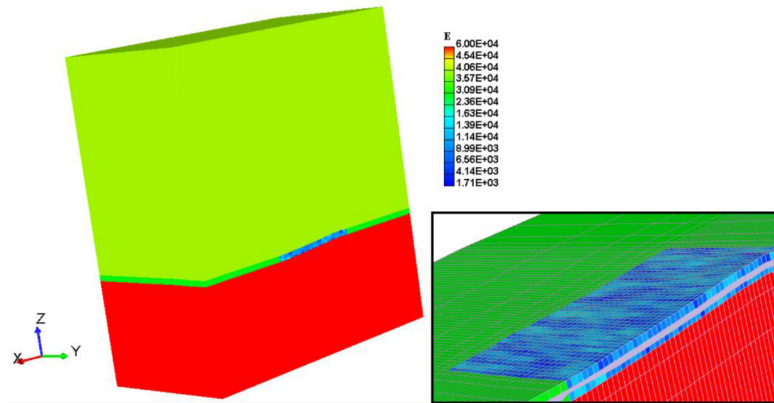


Figure 6—It depicts the reconstructed mechanics' mesh for Cranfield.

We assume common gradients for the pore pressure and in-situ stresses, such as [1]:

$$\sigma_h = 0.9 \text{Psi} / ft; \sigma_H = 1.0 \text{Psi} / ft; \sigma_v = 1.1 \text{Psi} / ft; p_0 = 0.445 \text{Psi} / ft; \rho = 5.4191 \cdot 10^{-5} \cdot z + 0.6503 \quad (4)$$

i.e., a normal faulting regime that is not self-equilibrating on purpose. We discretize the above gradients as evenly spaced in depth to vary the number of unknowns the optimizer needs to invert. We perturb the gradients in Eqn. (4) randomly. We want to optimize the density profile and the horizontal in-situ stresses simultaneously. There are 960 surface points for measurements, and we fixed the number of discrete points to six, which renders 18 unknown parameters ($n = N_w = 18$) [1].

Table 1 summarizes the optimization results for the two scenarios. Notice that BV-GN could not converge for this example. It struggled to reduce the objective below 279.7. The initial objective value for this case was about 288.69. Conversely, the DS-GN solved the problem in 12 iterations after 14 flops. ScyPy nonlinear least squares subroutine reported a first-order optimality of $3.14\text{e-}07$. For the heterogeneous case, we did not try BV-GN. DS-GN performed similarly to the layered case but needed two additional iterations to achieve similar stopping criteria. Conjugate gradient (CG), the linear solver to solve the elasticity, spent an average of ten more iterations per optimization flop.

Table 1—Coarse Cranfield problem ($n = 18$; $m = 960$; both ρ , σ_H , and σ_h are unknown).

Optimizer	Iter.	#Obj.	Objective
Layered			
BV-GN	9	12	279.7
DS-GN	12	14	0.06407
Heterogenous			
DS-GN	14	16	0.07178

Figure 7 shows the evolution of the vertical displacement, u_z , during the optimization process at relevant levels for the layered DS-GN run in Table 9. Similarly, Figure 8 shows the evolution of the mean stress, σ_m , during the same process. The optimizer undershoots and overshoots, rendering an alternating and faster (preferred) convergence. The latter is evident from the variation of the vertical displacement. However, the evolution of the mean stress was more monotone. It is also clear that the clamped boundary condition at the bottom is intrusive.

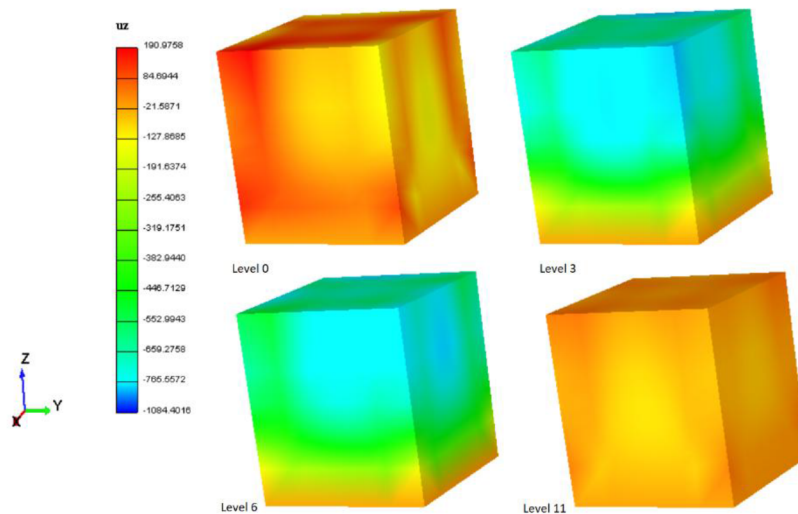


Figure 7—It depicts the evolution of the vertical displacement, u_z , for the Coarse Cranfield problem.

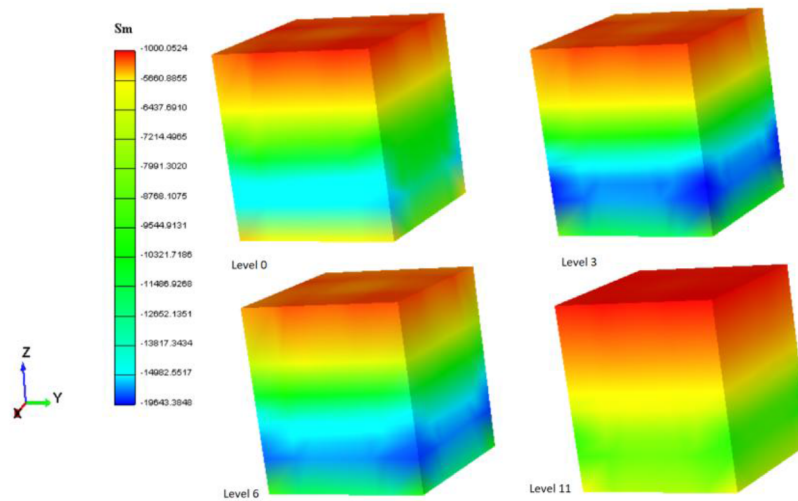


Figure 8—It depicts the evolution of the mean stress, σ_m , for the Coarse Cranfield problem.

Finally, Figures 9 and 10 depict vertical displacement, u_z , and the mean stress, σ_m , after 6, 12, and 18 years, respectively. Notice that we cut away half of the mechanics' domain. Figure 9 shows the classical compaction dome that grows from the pay-zone towards the overburden. A build-up also exists in the underburden but occupies a smaller area. Figure 10 highlights the equilibrated resulting mean stress and its transient variation in those years. We see significant stress variations within the reservoir due to the substantial depletion. Stresses are exceptionally high on the edges, but the overburden has no considerable arching. There are no significant differences between these magnitudes when we compare the two cases we tackled.

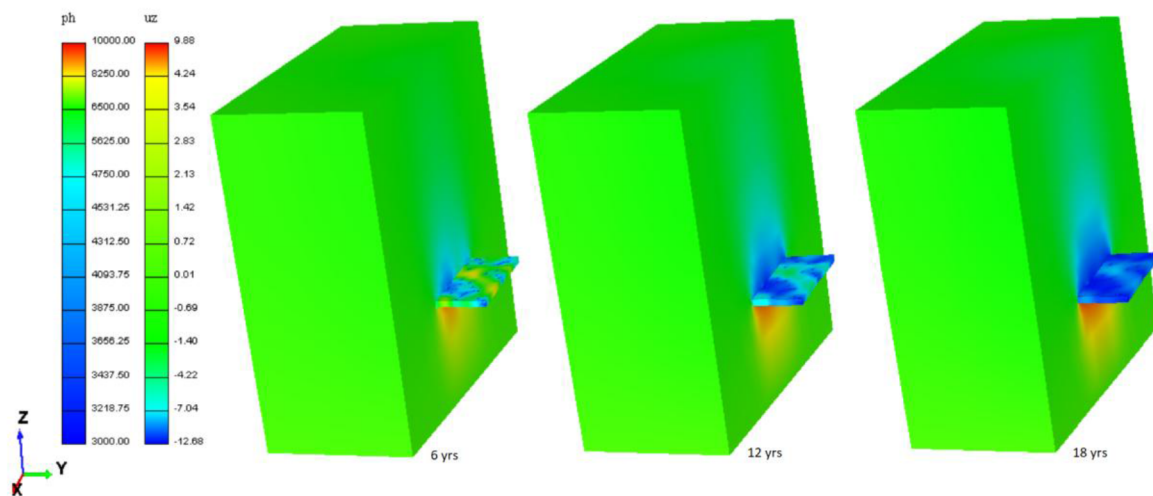


Figure 9—Sample u_z snapshots, for the Coarse Cranfield problem.

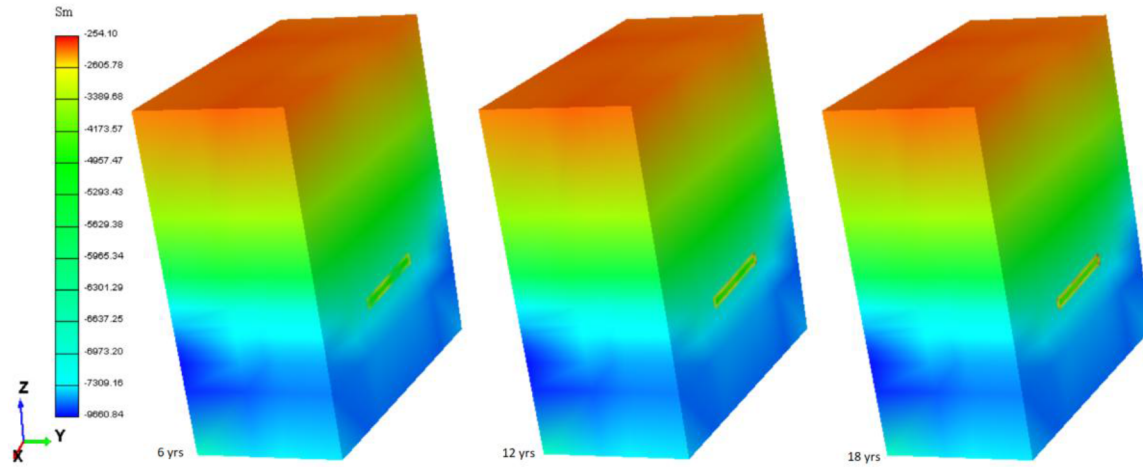


Figure 10—Sample σ_m snapshots, for the Coarse Cranfield problem.

Synthetic middle eastern giant reservoir

We also revisit a synthetic but realistic field-scale 16-million-cell flow model. The base flow grid is $256 \times 256 \times 256$ cells in size with 50 m areal spacing and 0.3 ft vertical spacing. The matrix porosity and permeability are heterogeneous, with permeability ranging from 20 mD to 800 mD and porosity from 15% to 25%. It is a two-phase gas-water model with gas dissolution in water. Figure 11 depicts the configuration of the four pads of 4 injector wells each. Target injection rates are 4.2 MTPA [17]. We generate a suitable geomechanics mesh by extending the smoothed-out reservoir grid into its surroundings. This is the classical tensor-product (TP) approach for such coupled models, as described above, see [12,13] for instance. The total number of grid blocks in the geomechanics model is 19.2 million, i.e., $402 \times 402 \times 100$ (see Figure 11).

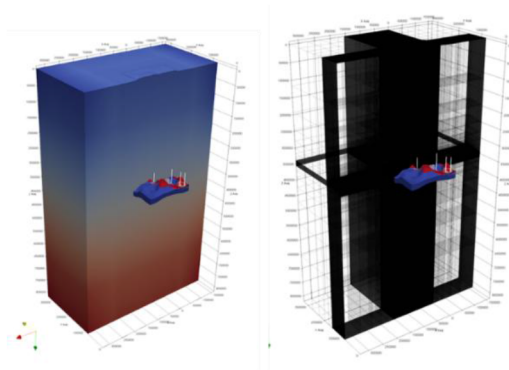


Figure 11—It depicts a gigantic synthetic reservoir after Cancelliere et al. [17].

We again borrow the gradients from the previous example, see Eq. (4), and perturb them randomly accordingly. We considered linear isotropic elasticity with mechanical properties for the reservoir's surroundings given by $E_o = 60$ ksi and $\nu_o = 0.3$. We aim to optimize the rock density profile and the horizontal in-situ stresses simultaneously. We remark that our in-house geomechanics code currently only supports linear variations of these quantities [8]. There are 162409 surface points for measurements and six unknown parameters, respectively. We also want to assess the role of the heterogeneity in the reservoir; thus, we consider porosity-based mechanical properties as [14]:

$$E = E_o \cdot \left(1.0 - \frac{\phi}{0.41}\right)^{1.19} ; \nu = \nu_p \cdot (\nu_o + 1.0) \left(\frac{\phi}{0.41}\right)^{0.0087}, \quad (5)$$

Table 2 summarizes the results. We performed the optimization runs in our in-house cluster and utilized 1024 processors per job. For the DS-GN run, the optimizer spent 17 function evaluations, the initial objective was $1.3264\text{e}+06$, and the final one was $1.5752\text{e}+01$, which renders a first-order optimality $2.49\text{e}+03$. The average runtime per objective function evaluation was about ≈ 31 minutes. Since the number of unknown parameters is small, both optimizers are competitive. However, the DS-GN reduces the objective further for the same iteration level and number of flops. We did not observe a significant challenge with the heterogeneity in the pay-zone, except that it increases the number of iterations of the iterative linear solver to solve every realization.

Table 2—Middle eastern giant reservoir problem ($n = 6$; $m = 162409$; both ρ , σ_H , and σ_h are unknown).

Optimizer	Iter.	#Obj.	Objective
$n = N_w = 6$			
BV-GN	19	23	$2.45\text{E}+01$
DS-GN	17	20	$1.57\text{E}+01$

Figure 12 shows the evolution of the mean stress, σ_m , during the optimization process at some iteration levels. As in the previous example, the optimizer undershoots and overshoots, rendering an alternating and faster convergence. The latter is evident from the variation of the objective function values during the process.

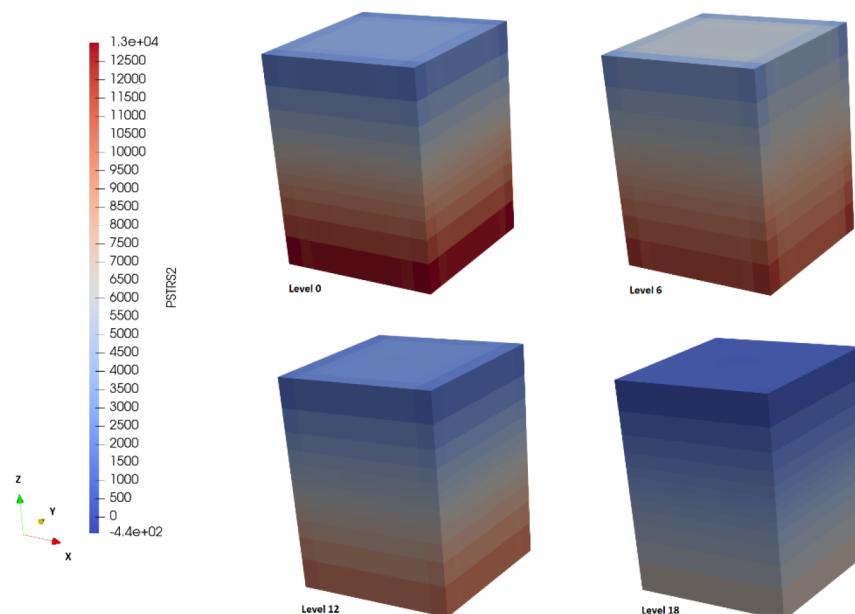


Figure 12—depicts the evolution of the mean stress, σ_m , for the Middle Eastern giant reservoir case.

Conclusions and path forward

We have presented herein comprehensive numerical results encompassing a whole 3-D synthetic field heterogeneous poroelastic case, one-way coupled with our in-house reservoir simulator, emphasizing the potential for CCS applications. We also tackled a gigantic synthetic Middle Eastern reservoir to show the approach's scalability on cluster computers. The results dictate that our optimization-based approach and evaluator's architecture can handle realistic stress fields that match the predefined measurements with a minimal computational cost regarding iterations and simulation jobs.

Our implementation of the general mechanical property and in-situ stresses evaluator is well encapsulated and maintainable. It can easily be updated to support other evaluation methods besides those already included in the enumeration. The resulting C++ source code is robust enough for all the use cases reported since the implementation is well tested and covered by dozens of unit tests.

We expect to parallelize our Python optimizer to fully exploit our cluster capabilities and thus reduce runtime. To improve robustness, we will handle more general use cases involving job failures. We may aim to extend the approach to simultaneously history-match flow and mechanical measurements.

References

1. Florez, H. and Cancelliere, M., 2025, March. Implementing a Duchon Spline Based on Gauss-Newton Distributed Optimizer for Coupled Flow and Geomechanics. In *SPE Reservoir Simulation Conference* (SPE-223866-MS).
2. Alpak, F., Gao, G., Florez, H., et al, 2023. A machine-learning-accelerated distributed LBFGS method for field development optimization: algorithm, validation, and applications. *Computational Geosciences*, **27**(3), 425–450.
3. Gao, G., Florez, H., Jost, Sean, et al, 2024. Implementation of Asynchronous Distributed Gauss-Newton Optimization Algorithms for Uncertainty Quantification by Conditioning to Production Data. *SPE Journal*, **29**(03), 1451–1472.
4. Gao, G., Florez, H., Jost, S., et al, 2022, September. Implementation of Asynchronous Distributed Gauss-Newton Optimization Algorithms for Uncertainty Quantification by Conditioning to Production Data. In *SPE Annual Technical Conference and Exhibition*.
5. Shahriar, M.F. and Khanal, A., 2023. Fundamental investigation of reactive-convective transport: Implications for long-term carbon dioxide (CO₂) sequestration. *International Journal of Greenhouse Gas Control*, **127**, p. 1–36.
6. Liu, P., Zhang, T. and Sun, S., 2019. A tutorial review of reactive transport modeling and risk assessment for geologic CO₂ sequestration. *Computers & geosciences*, **127**, pp.1–11.
7. Dogru, A.H., Fung, L.S., Middya, U., Al-Shaalan, T.M., Pita, J.A., Hemanth, Kumar, K. Su, Tan, J.C., Hoy, H., Dreiman, W.T. and Hahn, W.A., 2009, October. A next-generation parallel reservoir simulator for giant reservoirs. In *SPE/EAGE Reservoir Characterization & Simulation Conference* (pp. cp-170). European Association of Geoscientists & Engineers.
8. Qiao, L., Shen, M. and Dogru, A., 2024, February. Coupled Geomechanics and Reservoir Simulation for Gigantic Naturally Fractured Reservoirs. In *International Petroleum Technology Conference* (p. D031S138R008). IPTC.
9. Florez, H., Wheeler, M., Rodriguez, A.A., and Monteagudo, J.E., 2011, February. Domain decomposition methods applied to coupled flow-geomechanics reservoir simulation. In *SPE Reservoir Simulation Conference* (SPE-141596).
10. Florez, H., Wheeler, M.F., and Rodriguez, A. 2012. A mortar method based on NURBS for curved interfaces. *Proceedings of the 13th European Conference on the Mathematics of Oil Recovery* (ECMOR XIII).
11. Florez, H.A. 2012. Domain Decomposition Methods for Geomechanics. Ph.D. thesis, The University of Texas at Austin.
12. Florez, H.A., and Gildin, E. 2019. Model-order reduction of coupled flow and geomechanics in Ultra-Low Permeability (ULP) Reservoirs. *SPE Reservoir Simulation Conference*, (SPE-193911).
13. Florez, H. and Wheeler, M.F. 2016. A mortar method based on NURBS for curved interfaces. *Comput. Methods Appl. Mech. Engrg.*, **310**, pp. 535–566.
14. Kováčik, J. 1999. Correlation between Young's modulus and porosity in porous materials. *Journal of Materials Science Letters*, **18**(13), 1007–1010.

15. Florez, H., Manzanilla-Morillo, R., Florez, J., and Wheeler, M.F., 2014. Spline-based reservoir's geometry reconstruction and mesh generation for coupled flow and mechanics simulation. *Computational Geosciences*, **18**(6), 949–967.
16. Florez, H., and Ceberio, M. 2016. A Novel Mesh Generation Algorithm for Field-Level Coupled Flow and Geomechanics Simulations. In: *50th US Rock Mechanics/Geomechanics Symposium*, Houston. American Rock Mechanics Association.
17. Cancelliere, M., Florez, H.A., Al-Shaalan, T.M., & Kovalevskiy, O. 2024. Harnessing High-Performance Computing for Comprehensive Visualization of Large-Scale CO2 Sequestration Flow and Geomechanics Numerical Models. In: *SPE Europec featured at EAGE Conference and Exhibition*.